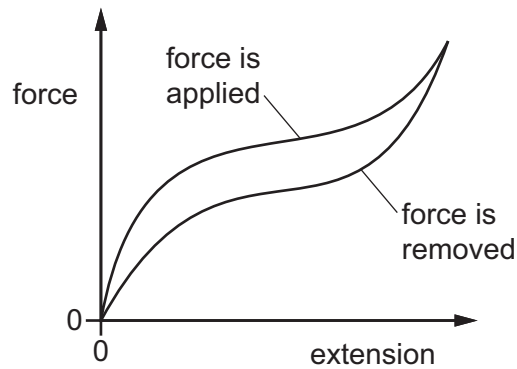


- 21 The graph shows the variation in extension with force for a sample of rubber.



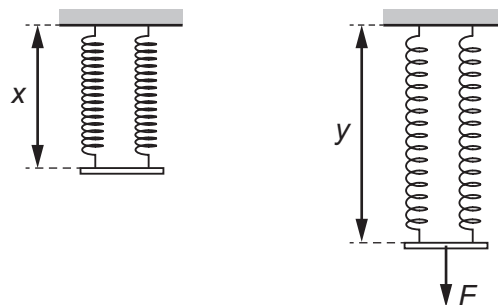
The top line shows the variation in extension as a force is applied.

The bottom line shows the variation in extension as the force is removed.

What is represented by the area between the two lines?

- A the work done as the force is applied
 - B the work done as the force is applied minus the work done as the force is removed
 - C the work done as the force is removed
 - D the work done as the force is removed plus the work done as the force is applied
- 22 Two identical springs have the same spring constant k . The springs are connected in parallel. The length of the unstretched springs is x .

A force F is applied to the spring combination. The length of the springs is now y .



Both springs are deformed within their limits of proportionality.

Which expression gives the elastic potential energy stored in **one** of the springs?

- A $\frac{1}{4}F(y - x)$
 - B $\frac{1}{2}F(y - x)$
 - C $\frac{1}{4}k(y - x)^2$
 - D $k(y - x)^2$
- 23 Which expression gives the formula for the spring constant?
- A Fx
 - B $2Fx^2$
 - C $\frac{F}{x}$
 - D $\frac{F}{x^2}$